BLUESTOCK MUTUAL FUND

ANALYTICS CAPSTONE PROJECT

B.Tech Artificial Intelligence and Data Science

Bluestock Internship Program

June 2026

Submitted By:

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Executive Summary:

This project focuses on the analysis of the Indian Mutual Fund industry using Python, SQL, Jupyter Notebook, and Power BI. The objective was to build a complete analytics solution covering data ingestion, cleaning, exploratory analysis, fund performance evaluation, risk analytics, investor behavior analysis, and interactive dashboard development.

The project utilized multiple mutual fund datasets containing fund information, NAV history, investor transactions, AUM statistics, portfolio holdings, and benchmark index data. Advanced analytics such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, Value at Risk (VaR), Conditional Value at Risk (CVaR), and fund recommendation systems were implemented.

The final outcome includes a Power BI dashboard, analytical reports, and actionable insights for mutual fund investors and analysts.

Problem Statement & Objectives:

Problem Statement:

Mutual fund investors require comprehensive insights into fund performance, risk, investor behavior, and market trends. Raw financial data alone does not provide actionable information. Therefore, an end-to-end analytics platform is required to transform raw data into meaningful insights.

Objectives:

- **Build ETL pipeline for mutual fund datasets**
- **Perform data cleaning and validation**
- **Conduct exploratory data analysis**
- **Compute fund performance metrics**
- **Analyze investor behavior**
- **Develop risk analytics models**
- **Create interactive Power BI dashboard**
- **Generate recommendations and insights**

Data Sources:

Dataset	Description
Fund Master	Scheme metadata
NAV History	Daily NAV values
AUM Data	Assets under Management
SIP Inflows	Monthly SIP investments
Category Inflows	Category-wise investments
Folio Count	Investor folio statistics
Scheme Performance	Return and risk metrics
Investor Transactions	SIP and redemption data
Portfolio Holdings	Fund portfolio allocation
Benchmark Indices	NIFTY benchmark data

ETL Architecture:

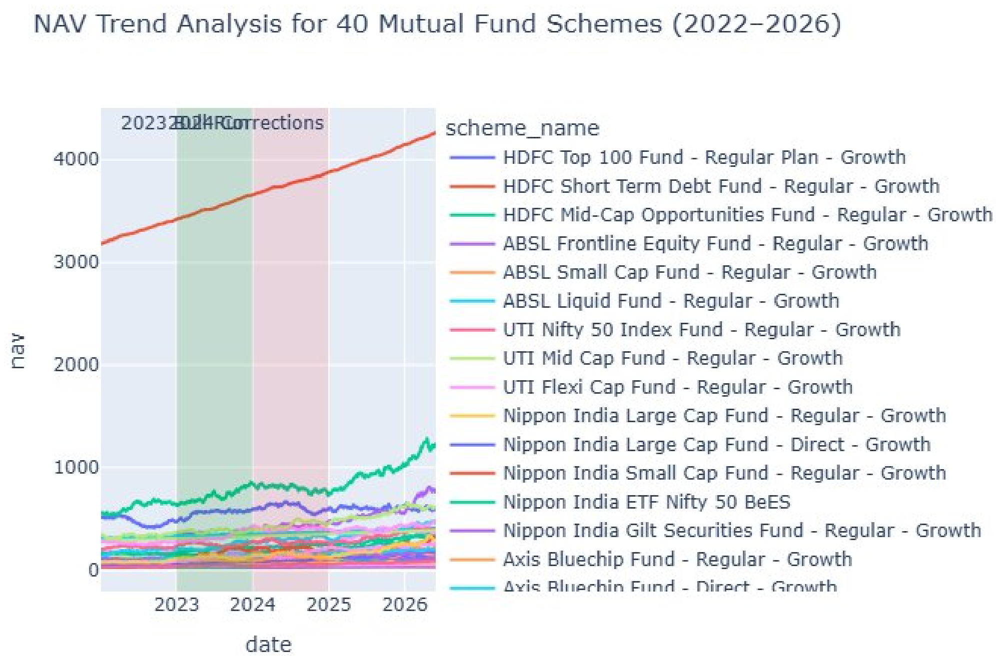
The ETL pipeline was designed using Python and Pandas. Raw CSV datasets were ingested, validated, cleaned, transformed, and loaded into a SQLite database for analytics processing.

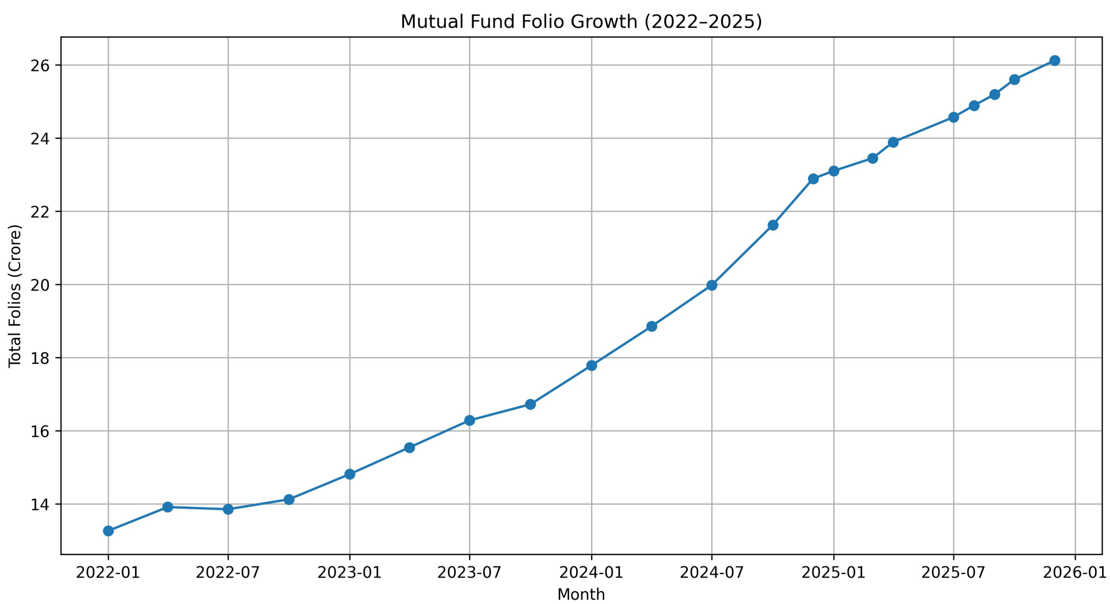
Data Cleaning Process:

- Date standardization
- Missing value handling
- Duplicate removal
- Data type validation
- NAV validation
- Transaction validation
- Expense ratio validation
- AMFI code verification

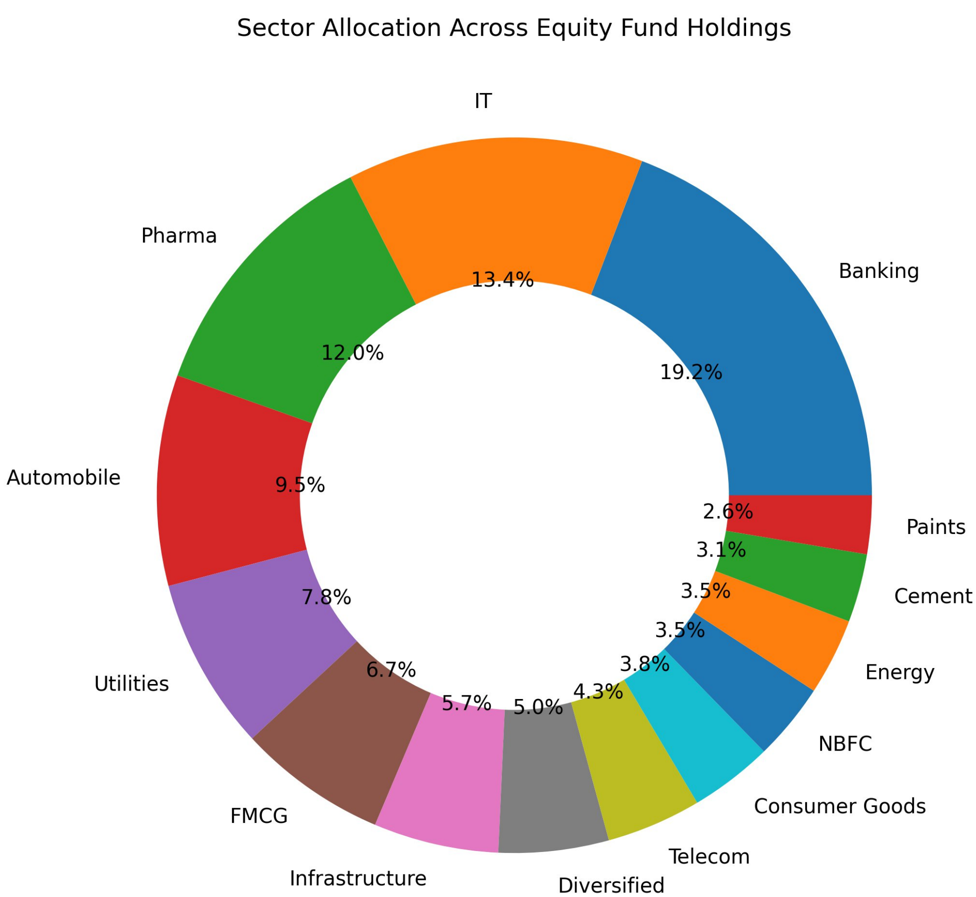
EDA Findings:

1. SIP inflows increased consistently from 2022–2025.
2. SBI Mutual Fund maintained market leadership.
3. Equity categories attracted the highest inflows.
4. Industry folio count showed strong growth.
5. Small-cap funds delivered superior returns.

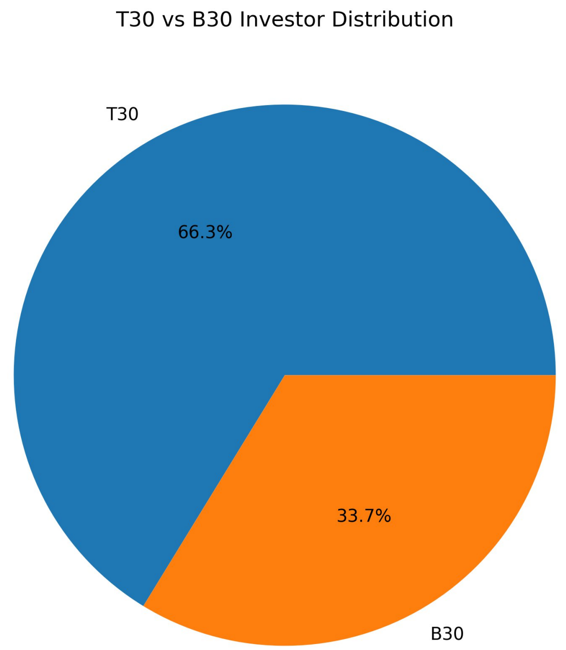

NAV Trend Analysis for 40 Mutual Fund Schemes (2022–2026)
2023 Bull Run
2024 Corrections
nav
0
1000
2000
3000
4000
2023
2024
2025
2026
date
scheme_name
HDFC Top 100 Fund - Regular Plan - Growth
HDFC Short Term Debt Fund - Regular - Growth
HDFC Mid-Cap Opportunities Fund - Regular - Growth
ABSL Frontline Equity Fund - Regular - Growth
ABSL Small Cap Fund - Regular - Growth
ABSL Liquid Fund - Regular - Growth
UTI Nifty 50 Index Fund - Regular - Growth
UTI Mid Cap Fund - Regular - Growth
UTI Flexi Cap Fund - Regular - Growth
Nippon India Large Cap Fund - Regular - Growth
Nippon India Large Cap Fund - Direct - Growth
Nippon India Small Cap Fund - Regular - Growth
Nippon India ETF Nifty 50 BeES
Nippon India Gilt Securities Fund - Regular - Growth
Axis Bluechip Fund - Regular - Growth
Axis Bluechip Fund - Direct - Growth

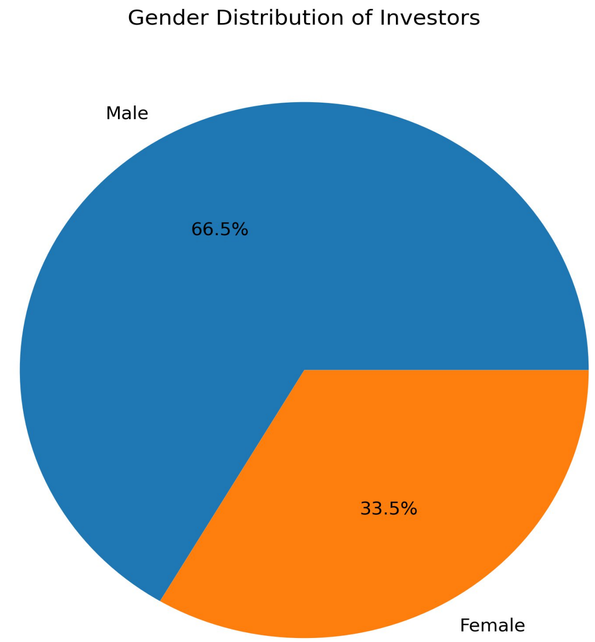

Mutual Fund Folio Growth (2022–2025)
Total Folios (Crore)
14
16
18
20
22
24
26
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
Month

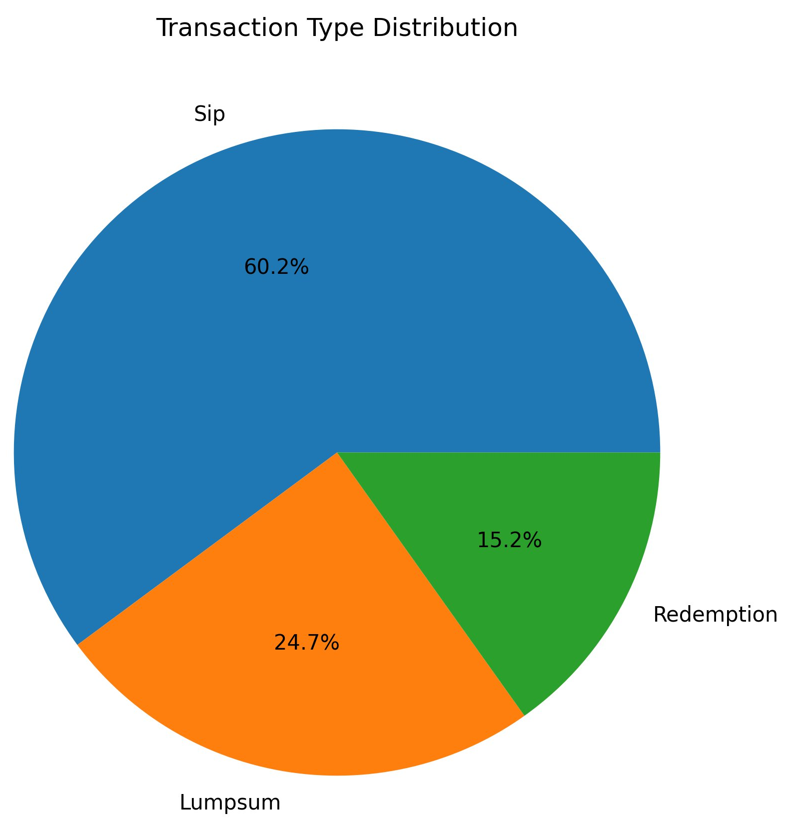

Monthly SIP Inflow Trend (2022–2025)
All-Time High: ₹31,002 Cr
sip_inflow_crore
30k
25k
20k
15k
10k
2022
2023
2024
2025
2026
month

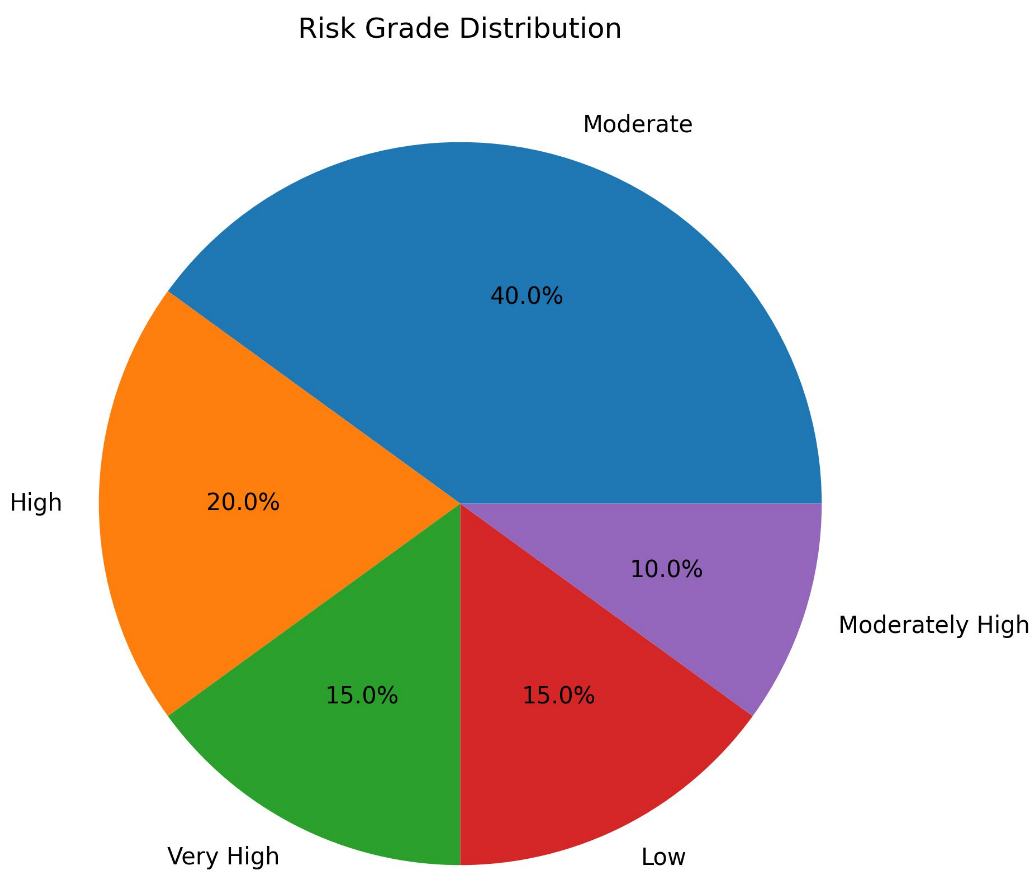

Sector Allocation Across Equity Fund Holdings
IT
Pharma
Banking
Automobile
Utilities
FMCG
Infrastructure
Diversified
Telecom
Consumer Goods
NBFC
Energy
Cement
Paints
19.2%
13.4%
12.0%
9.5%
7.8%
6.7%
5.7%
5.0%
4.3%
3.8%
3.5%
3.5%
3.1%
2.6%

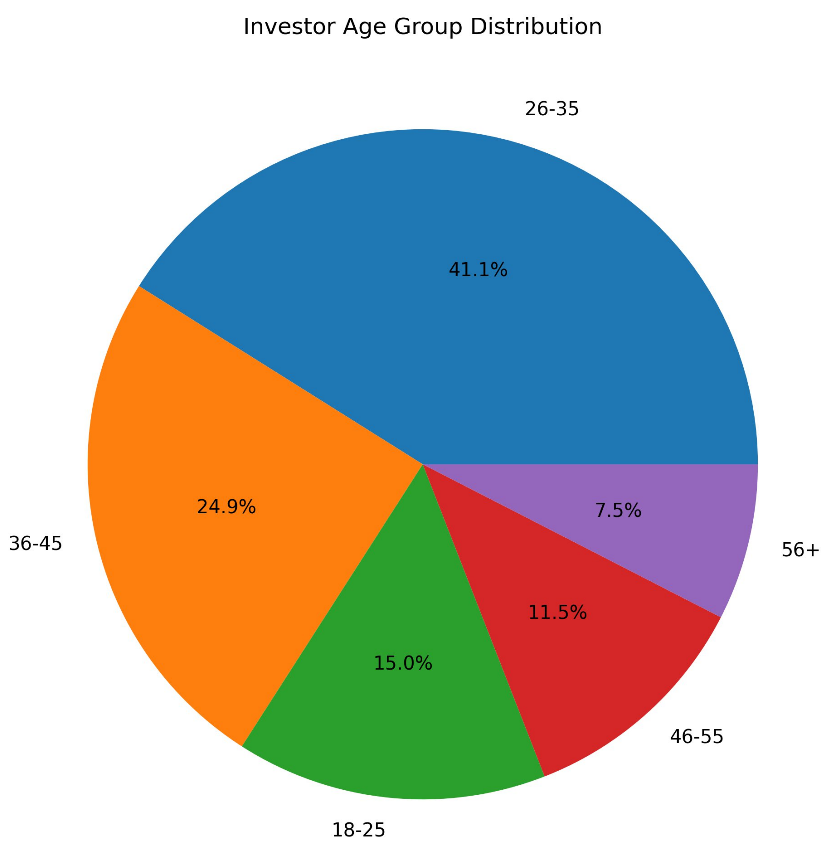
Distributions:








Risk Grade Distribution
Moderate
40.0%
High
20.0%
15.0%
15.0%
10.0%
Moderately High
Very High
Low


Investor Age Group Distribution
26-35
41.1%
36-45
24.9%
7.5%
56+
11.5%
15.0%
46-55
18-25

Performance Analytics:

Introduction

Performance analytics was performed to evaluate the effectiveness of mutual fund schemes using historical NAV data. The objective was to identify funds that generated superior returns while maintaining acceptable risk levels.

The analysis was conducted on 40 mutual fund schemes across multiple categories using historical NAV records, benchmark indices, and scheme performance metrics.

Daily Return Calculation

Daily returns were calculated using historical NAV values.

Formula:

Daily Return = $(NAV_t / NAV_{t-1}) - 1$

Daily return data serves as the foundation for performance and risk calculations.

Compound Annual Growth Rate (CAGR)

CAGR measures the annualized growth rate of an investment over a specified period.

Formula:

CAGR = (Ending NAV / Beginning NAV)^(1/n) − 1

Where:

- Ending NAV = Final NAV value
- Beginning NAV = Initial NAV value
- n = Number of years

CAGR provides a standardized method for comparing long-term fund performance.

Key Observations

- Large-cap funds exhibited relatively stable growth.
- Small-cap funds delivered higher returns but with increased volatility.
- Direct plans generally outperformed regular plans due to lower expense ratios.
- Equity-oriented funds generated stronger long-term returns compared to debt-oriented funds.

Sharpe Ratio

The Sharpe Ratio measures excess return generated per unit of total risk.

Formula: Sharpe Ratio = $(R_p - R_f) / \sigma_p$

Where: R_p = Portfolio Return R_f = Risk-Free Rate σ_p = Portfolio Standard Deviation

Funds with higher Sharpe Ratios provided better risk-adjusted returns.

Sortino Ratio

The Sortino Ratio focuses only on downside risk.

Formula:

Sortino Ratio = (Rp − Rf) / Downside Deviation

This metric is particularly useful for investors concerned with negative volatility.

Alpha

Alpha measures the excess return generated beyond benchmark performance.

Interpretation:

- Positive Alpha → Outperformed benchmark
- Negative Alpha → Underperformed benchmark

Beta

Beta measures fund sensitivity relative to market movements.

Interpretation:

- Beta > 1 → More volatile than market
- Beta < 1 → Less volatile than market

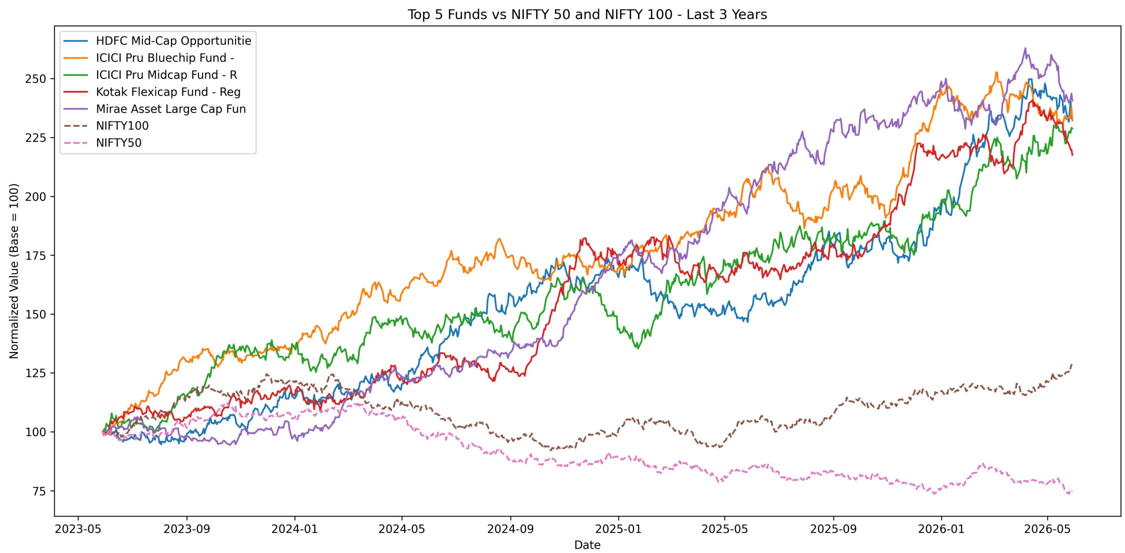
Maximum Drawdown

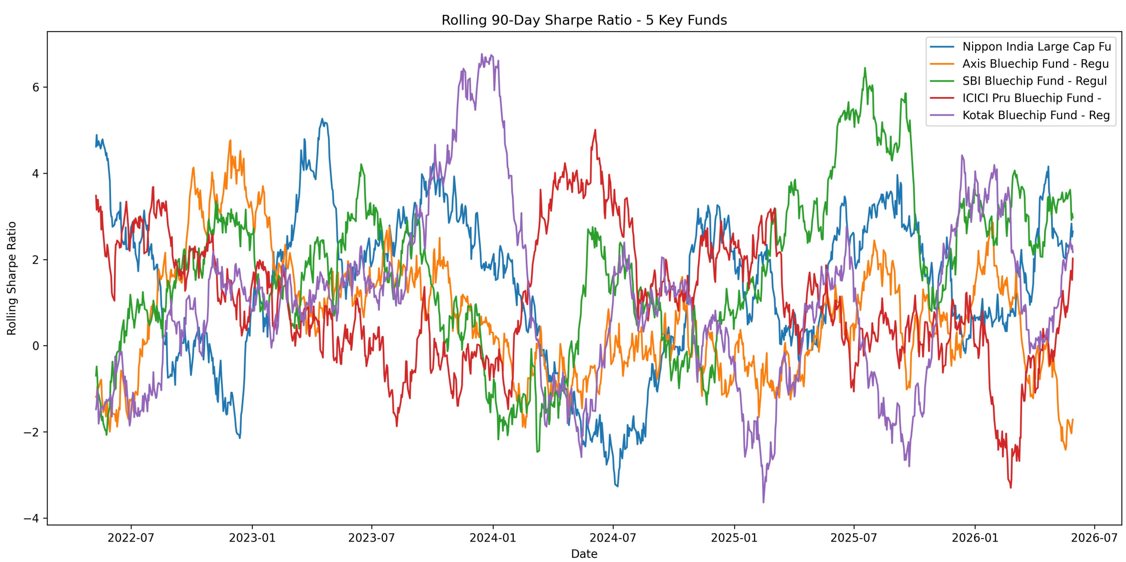
Maximum Drawdown represents the largest peak-to-trough decline experienced by a fund.

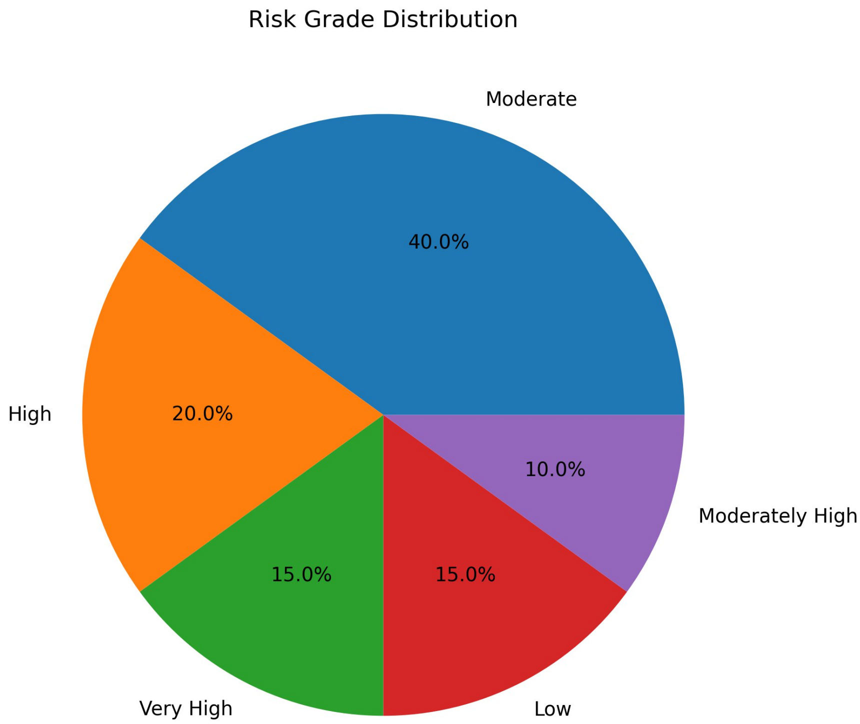
This metric helps investors understand the worst-case historical loss scenario.

Key Findings

- Several equity funds achieved consistently high Sharpe Ratios.
- Direct plans generally achieved better Alpha values.
- Low Beta funds demonstrated stronger resilience during market corrections.
- Maximum Drawdown varied significantly across categories.


Top 5 Funds vs NIFTY 50 and NIFTY 100 - Last 3 Years
HDFC Mid-Cap Opportunitie
ICICI Pru Bluechip Fund -
ICICI Pru Midcap Fund - R
Kotak Flexicap Fund - Reg
Mirae Asset Large Cap Fun
NIFTY100
NIFTY50
Normalized Value (Base = 100)
Date


Rolling 90-Day Sharpe Ratio - 5 Key Funds
Nippon India Large Cap Fu
Axis Bluechip Fund - Regu
SBI Bluechip Fund - Regul
ICICI Pru Bluechip Fund -
Kotak Bluechip Fund - Reg
Rolling Sharpe Ratio
Date



Dashboard Overview:

Power BI Dashboard

A four-page interactive Power BI dashboard was developed to enable business users and investors to explore mutual fund analytics.

Page 1 – Industry Overview

Provides a high-level view of the mutual fund industry including:

- Total AUM
- Total Schemes
- Investor Folios
- SIP Inflows
- Fund House Comparison

Page 2 – Fund Performance

Provides detailed performance insights including:

- Risk vs Return Analysis
- Fund Scorecard
- NAV Trends
- Fund Comparison

Page 3 – Investor Analytics

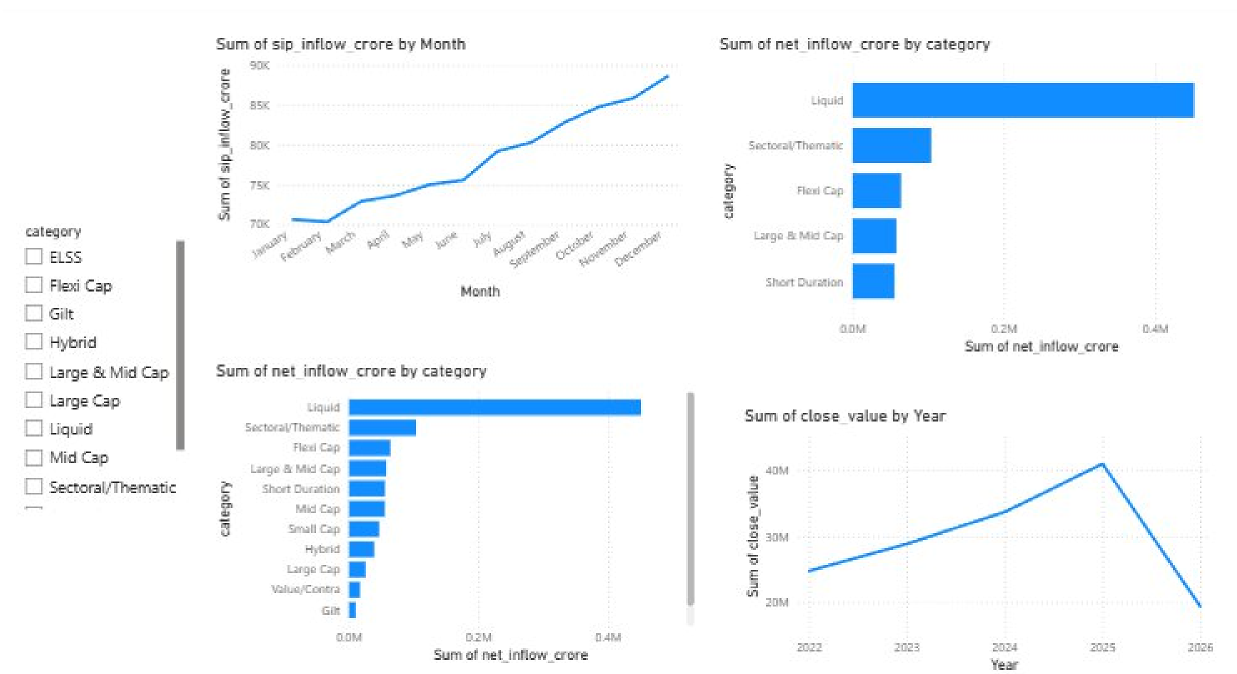
Provides insights into investor behavior:

- State-wise Investments
- Transaction Analysis
- Age Group Distribution
- SIP Patterns

Page 4 – SIP & Market Trends

Analyzes:

- SIP Growth Trends
- Category Inflows
- Market Movement
- Investment Preferences



fund_house

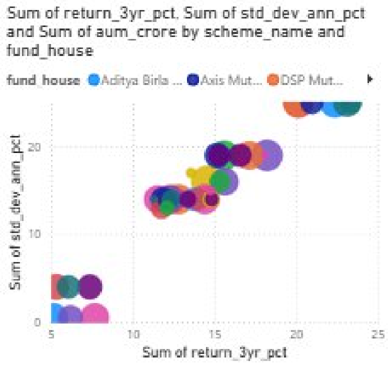
- Aditya Birla Sun Life MF
- Axis Mutual Fund
- DSP Mutual Fund
- HDFC Mutual Fund
- ICICI Prudential MF

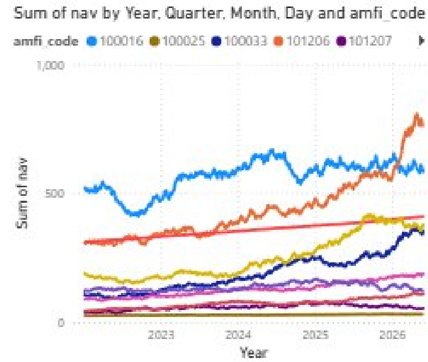
plan

- Direct
- Regular

category

- Debt
- Equity

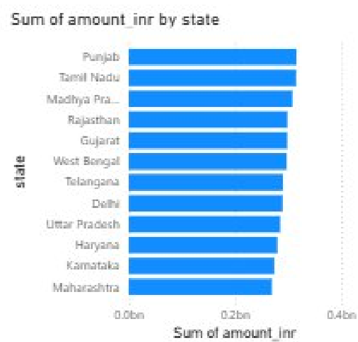


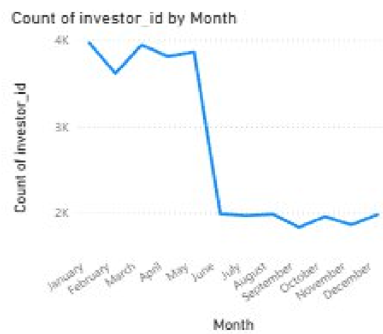


scheme_name	Sum of cagr_3yr_pct	Sum of final_score	Sum of sharpe_ratio	Sum of alpha_pct
Mirae Asset Large Cap Fund - Regular - Growth	34.00	86.25	1.45	26.98
ICICI Pru Midcap Fund - Regular - Growth	31.78	82.25	1.18	29.26
Kotak Flexicap Fund - Regular - Growth	29.58	82.00	1.31	27.33
HDFC Mid-Cap Opportunities Fund - Regular - Growth	32.44	80.75	1.09	27.20
ICICI Pru Bluechip Fund - Direct - Growth	32.49	80.00	1.03	21.19
Axis Midcap Fund - Regular - Growth	35.11	77.00	1.00	26.08
SBI Bluechip Fund - Regular Plan - Growth	30.46	74.81	1.21	23.20
Mirae Asset Tax Saver Fund - Regular - Growth	29.18	73.69	1.23	28.27
ABSL Frontline Equity Fund - Regular - Growth	28.97	68.19	1.03	21.40
SBI Small Cap Fund - Regular Plan - Growth	26.67	67.38	0.95	30.34
DSP Midcap Fund - Regular - Growth	26.87	66.75	1.13	26.60
Total	**656.59**	**2,050.00**	**21.49**	**636.34**

state

- Delhi
- Gujarat
- Haryana
- Karnataka
- Madhya Pradesh
- Maharashtra
- Punjab
- Rajasthan
- Tamil Nadu
- Telangana



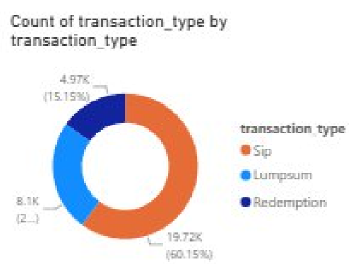


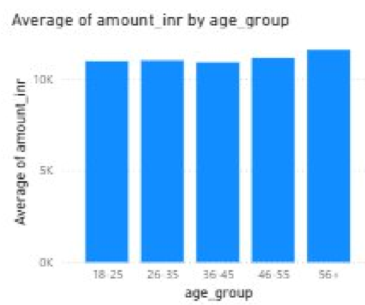
age_group

- 18-25
- 26-35
- 36-45
- 46-55
- 56+

city_tier

- B30
- T30





Mutual Fund Industry Overview Dashboard

Advanced Analytics:

Investors were grouped based on the year of their first transaction.

Objectives:

- Identify investor behavior trends.
- Compare investment activity across cohorts.
- Evaluate long-term engagement.

Findings

- Recent cohorts demonstrated higher average SIP contributions.
- Investor participation increased significantly over time.

SIP Continuity Analysis

SIP continuity measures investor consistency.

Investors with transaction gaps greater than 35 days were classified as:

At-Risk Investors

Benefits:

- Identifies churn risk.
- Enables proactive investor engagement.

Findings

- Most investors maintained regular SIP schedules.
- A small segment exhibited irregular contribution patterns.

Key Findings:

Major Insights

1. The Indian mutual fund industry experienced strong growth from 2022 to 2025.
2. Monthly SIP inflows reached record levels, reflecting growing investor confidence.
3. Equity-oriented schemes consistently attracted the highest inflows.
4. High Sharpe Ratio funds delivered superior risk-adjusted performance.
5. Small-cap schemes generated higher returns but carried greater downside risk.
6. Direct plans generally outperformed regular plans due to lower expenses.
7. Portfolio diversification significantly reduced concentration risk.
8. Investor participation increased steadily across all cohorts.
9. SIP continuity emerged as a strong indicator of long-term investment discipline.
10. Risk metrics provided valuable insights beyond traditional return analysis.

Limitations:

- Historical performance does not guarantee future returns.
- Dataset size and scope were limited.
- Some assumptions were used in risk calculations.
- Market conditions continuously evolve.

Recommendations:

- Investors should focus on risk-adjusted returns rather than absolute returns.
- SIP investing should be encouraged for long-term wealth creation.
- Portfolio diversification should be maintained.
- Regular performance monitoring should be conducted.
- Risk metrics should be incorporated into investment decision-making.

Conclusion:

This project successfully developed a comprehensive Mutual Fund Analytics platform integrating ETL, data engineering, statistical analysis, risk management, investor analytics, and business intelligence dashboards.

The solution demonstrates how data analytics can transform raw financial information into actionable insights for investors, analysts, and financial institutions. The project achieved all objectives and delivered a complete end-to-end analytics framework using Python, SQL, Jupyter Notebook, SQLite, and Power BI.